

Cloud Computing-Lab 01

Lab Number: Lab 01

Title: Ubuntu Server 24.04 LTS Installation & SSH Setup

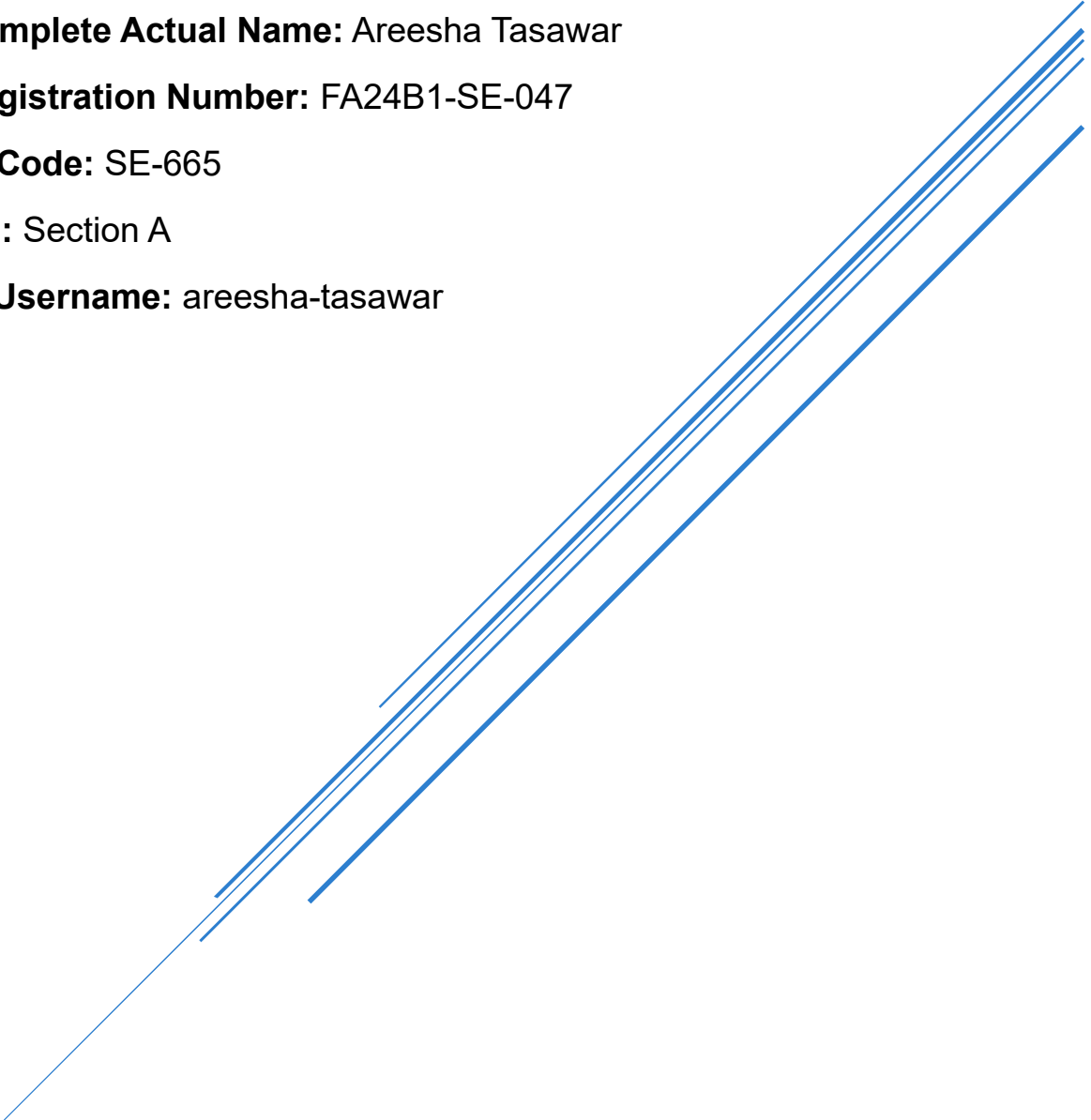
Your Complete Actual Name: Areesha Tasawar

Your Registration Number: FA24B1-SE-047

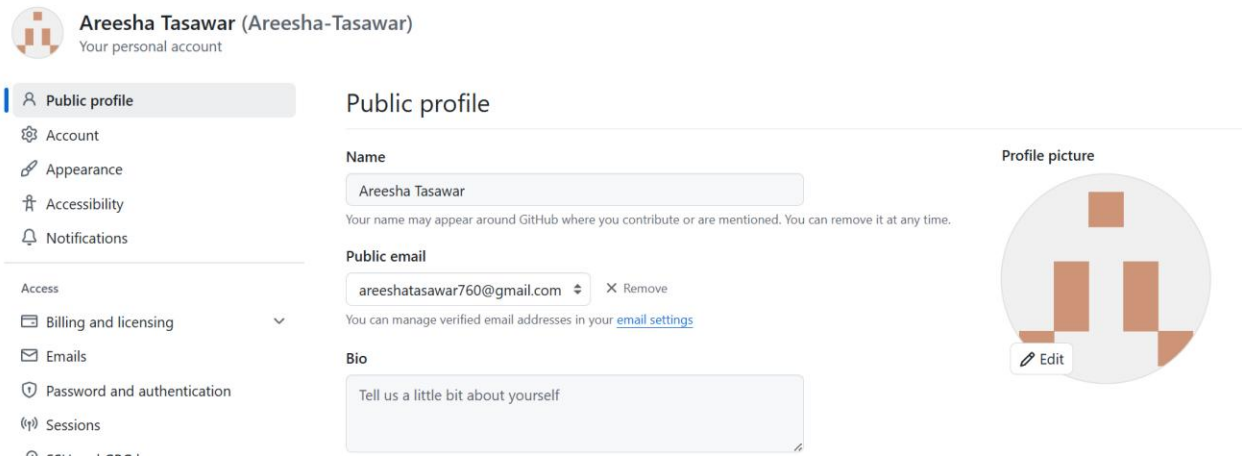
Course Code: SE-665

Section: Section A

GitHub Username: areesha-tasawar

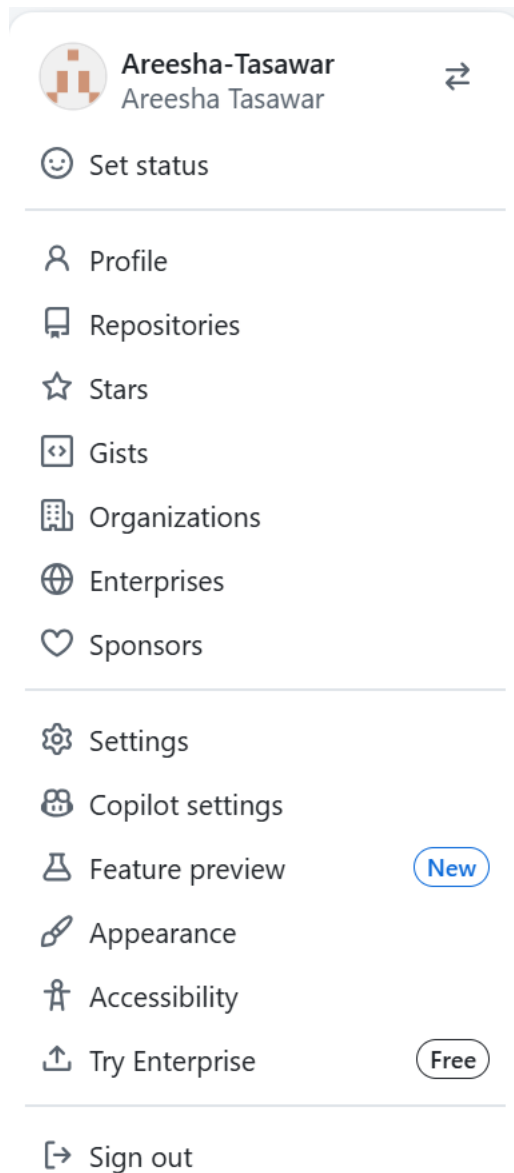
A series of five parallel blue lines of varying lengths, slanted diagonally from the bottom-left towards the top-right, located in the lower right quadrant of the page.

Task 1: Create and configure Your GitHub Account



Filename: github_profile.png

Figure 1: Public Github profile page showing profile picture, display name, and username.



Filename: github_actual_name.png

Figure 2: GitHub account profile name configured with full name.

Task 2: Register in the Student Marks Portal



Student Marks Portal

Sign up or sign in securely with your GitHub account.

Continue with GitHub

Filename: portal_login_page.png

Figure 3: Main Login page of the student Marks portal.



Complete registration

Enter your registration number, course and section. The enrollment requires approval.

Registration number

2024-BSE-47

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

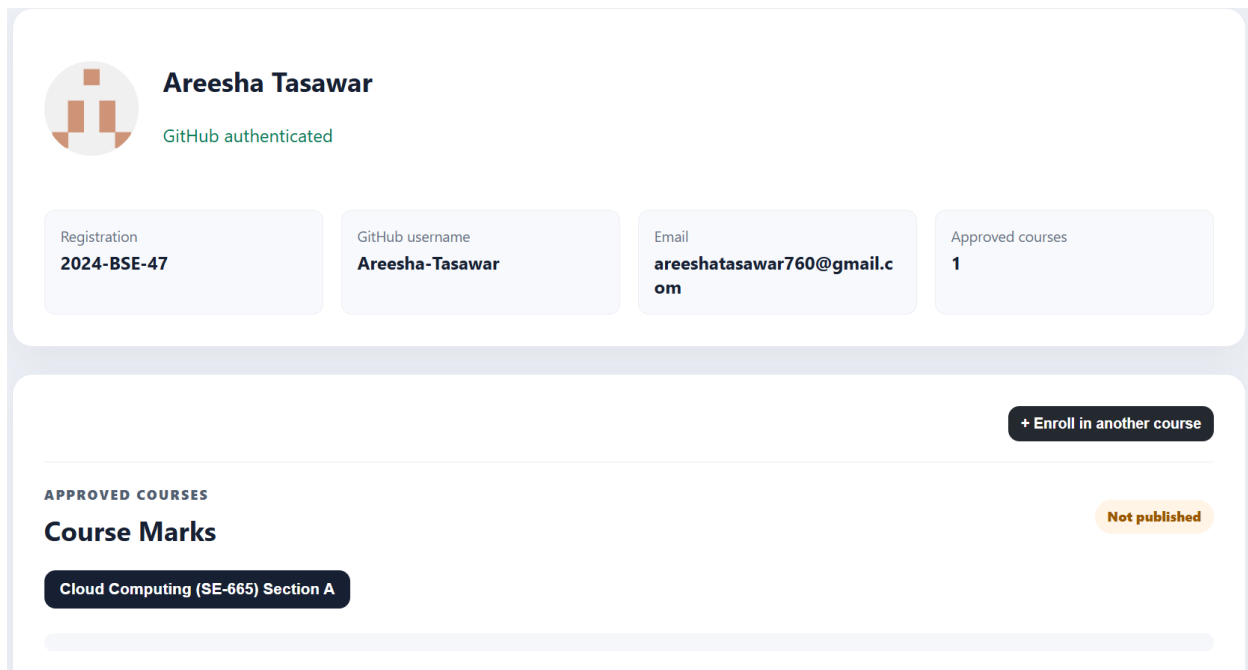
Section A

Submit enrollment

Sign out

Filename: portal_github_profile.png

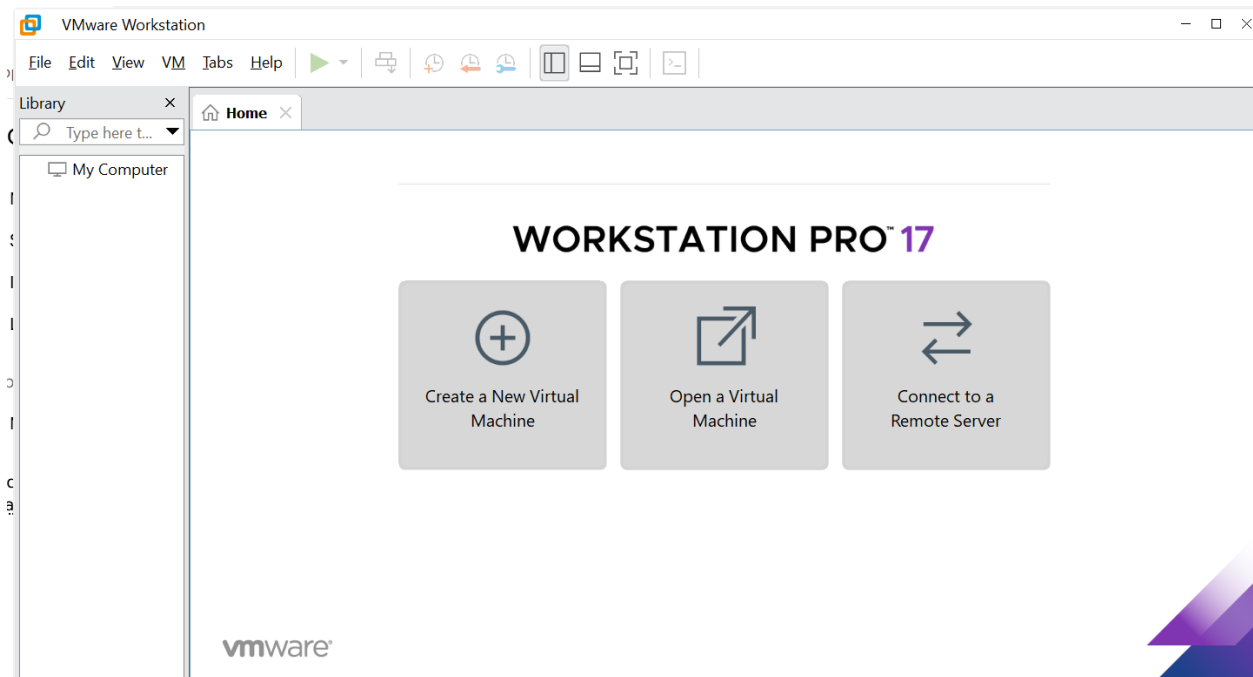
Figure 4: GitHub profile authentication status on Student Marks Portal.



Filename: portal_enrollment_submitted.png

Figure 5: Course enrollment confirmation for Cloud Computing Section A.

Task 3: Download VMware Workstation Pro and ubuntu Server



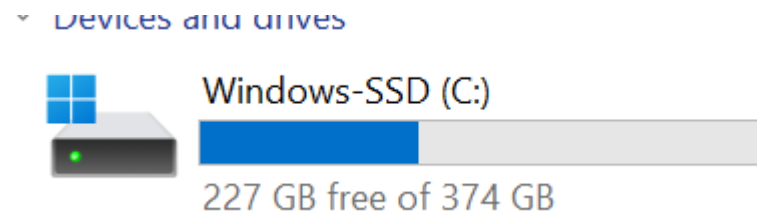
Filename: vmware_workstation.png

Figure 6: VMware Workstation Pro home screen setup.

ready			
 ubuntu-26.04.1-live-server-amd64.iso	9/12/2026 11:49 AM	Disc Image File	2,859,240 KB

Filename: ubuntu_server_iso.png

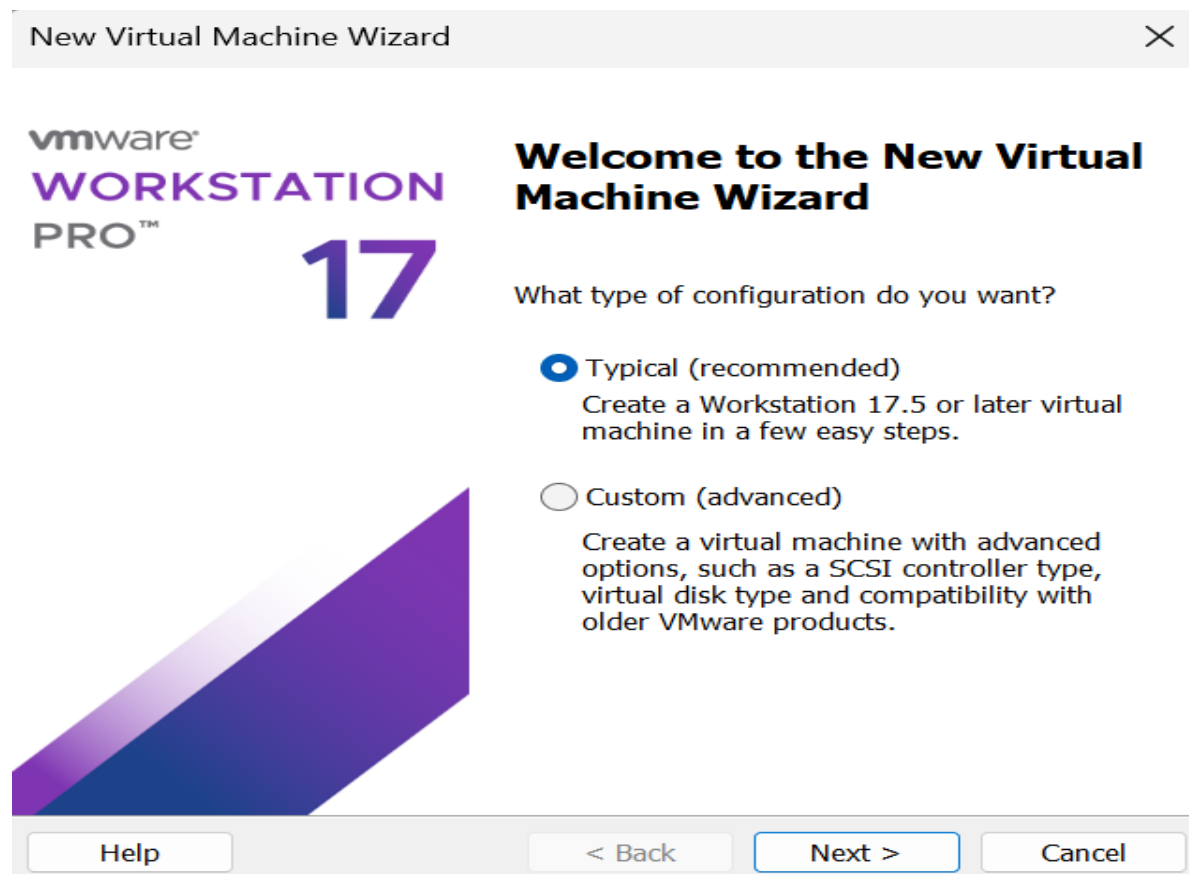
Figure 7: Downloaded Ubuntu Server ISO image file verification

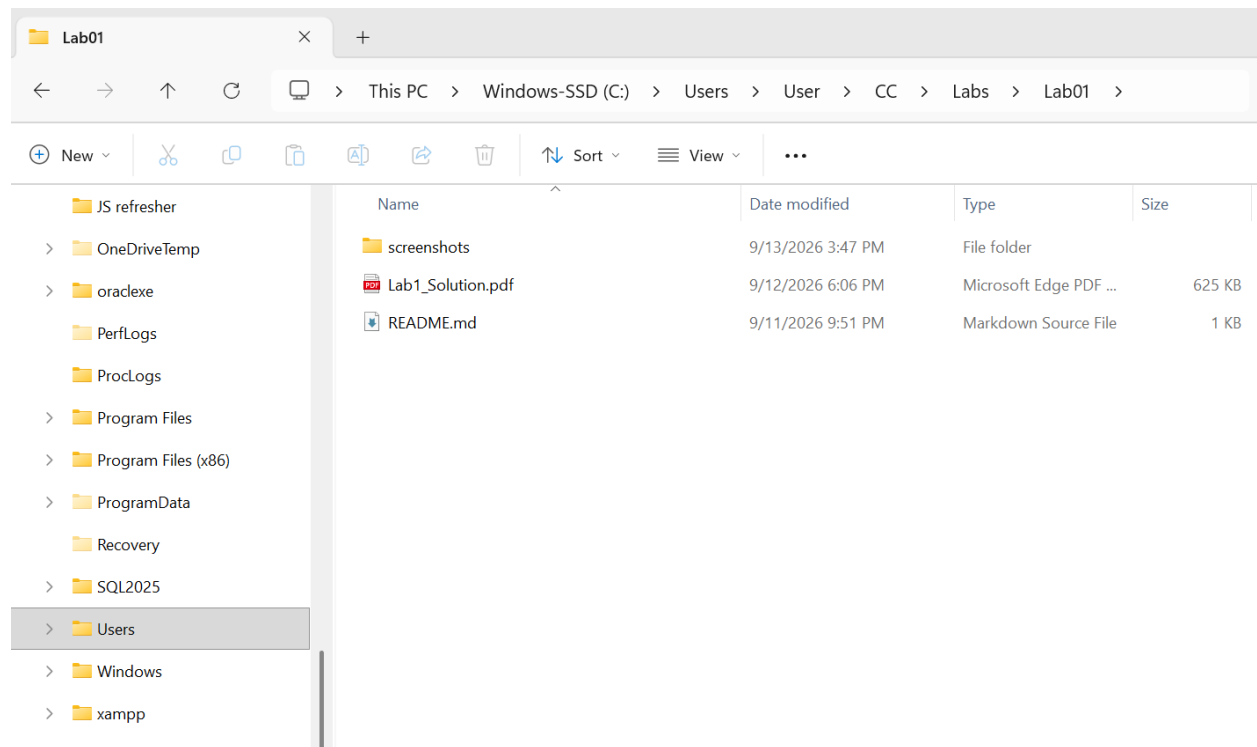


Filename: available_storage.png

Figure 8: Checking local drive storage space for Virtual Machine setup.

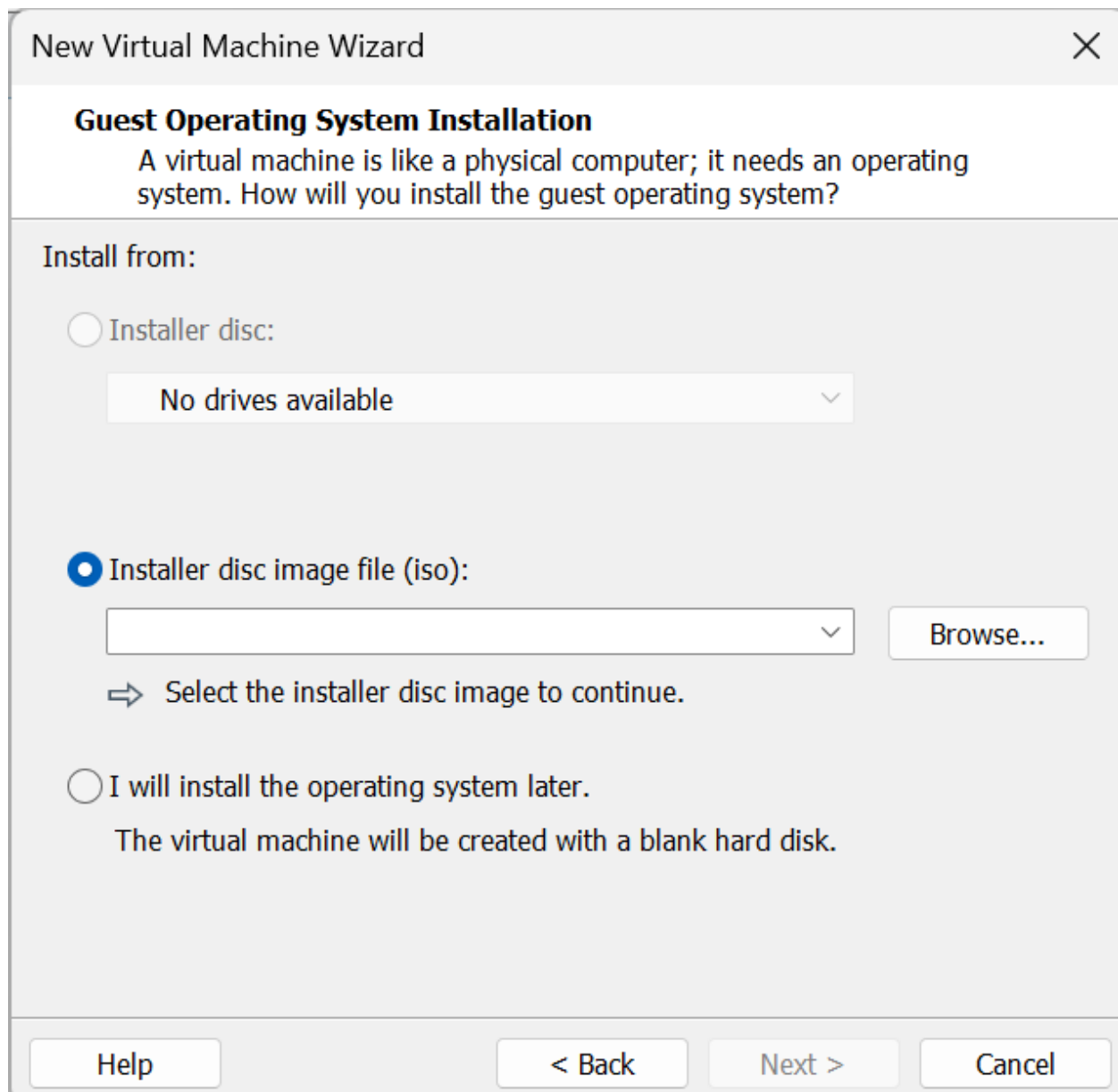
Task 4: Create the Ubuntu Server Virtual Machine





Filename:vm_creation_wizard.png

Figure 9:New virtual Machine Wizard interface in VMware Workstation.



The image shows a screenshot of the 'New Virtual Machine Wizard' window, specifically the 'Guest Operating System Installation' step. The window has a title bar with the text 'New Virtual Machine Wizard' and a close button (X) in the top right corner. Below the title bar, the section is titled 'Guest Operating System Installation' in bold. Underneath this title, a descriptive text reads: 'A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?'. The main area of the wizard is titled 'Install from:' and contains three radio button options. The first option is 'Installer disc:', which is currently unselected. Below it is a dropdown menu that displays 'No drives available'. The second option is 'Installer disc image file (iso):', which is selected with a blue radio button. Below this option is an empty text box with a dropdown arrow on its right side, and a 'Browse...' button to its right. Below the text box is a right-pointing arrow icon followed by the text 'Select the installer disc image to continue.'. The third option is 'I will install the operating system later.', which is unselected. Below this option is a line of text: 'The virtual machine will be created with a blank hard disk.'. At the bottom of the wizard, there are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'.

New Virtual Machine Wizard

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☒ Installer disc image file (iso):

Browse...

⇒ Select the installer disc image to continue.

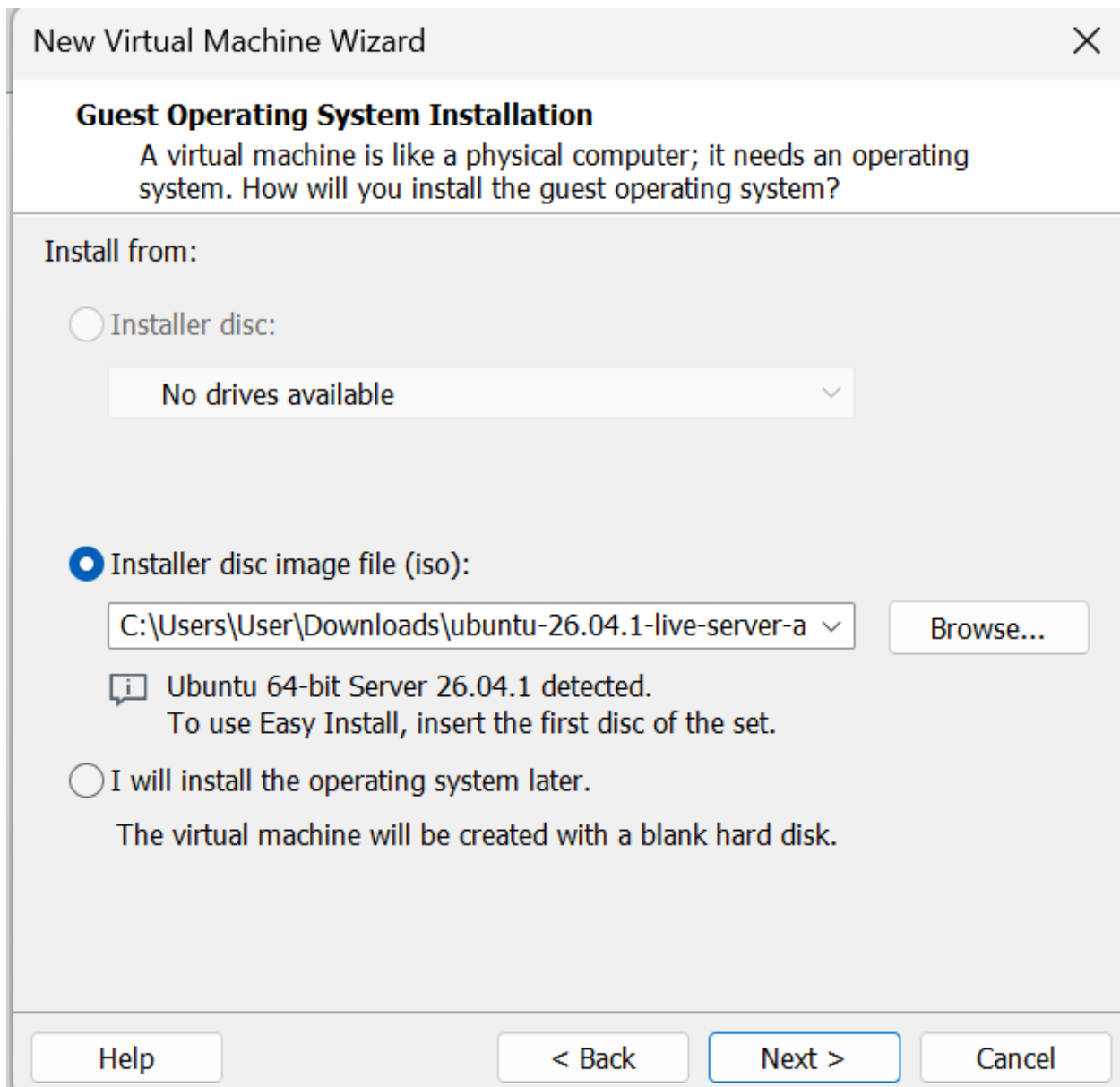
☐ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

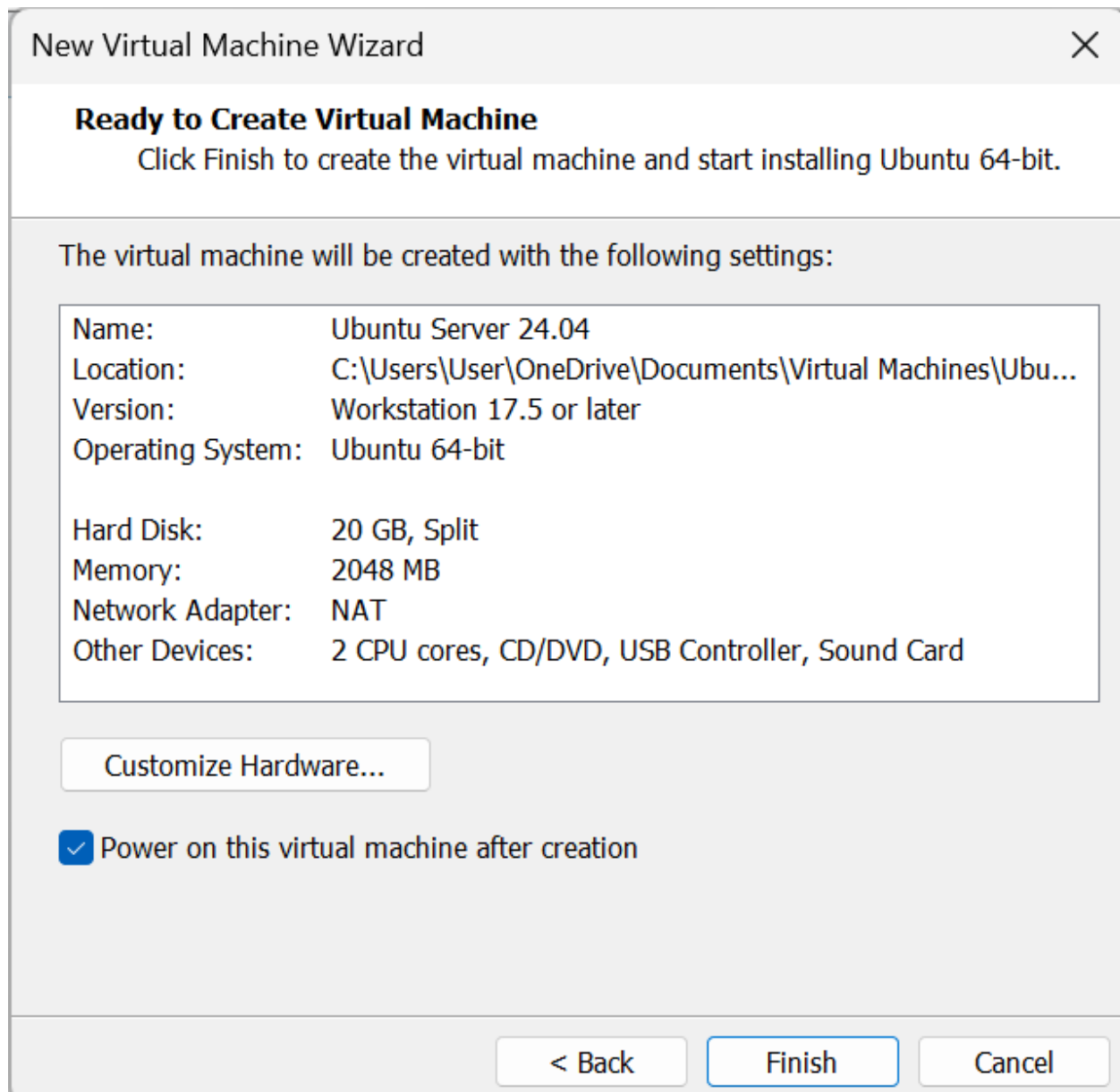
Filename: vm_typical_configuration.png

Figure 10: Selecting the Typical virtual machine configuration option.



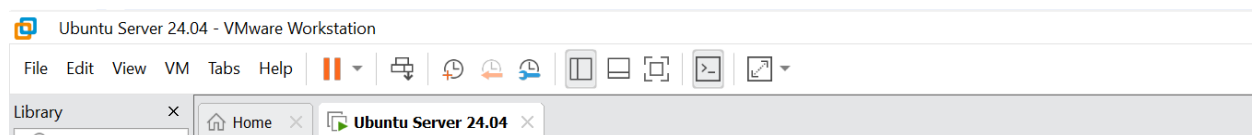
Filename: vm_typical_iso_selected.png

Figure 11: Browsing and selecting the downloaded Ubuntu server ISO path.



Filename:vm_configuration_summary.png

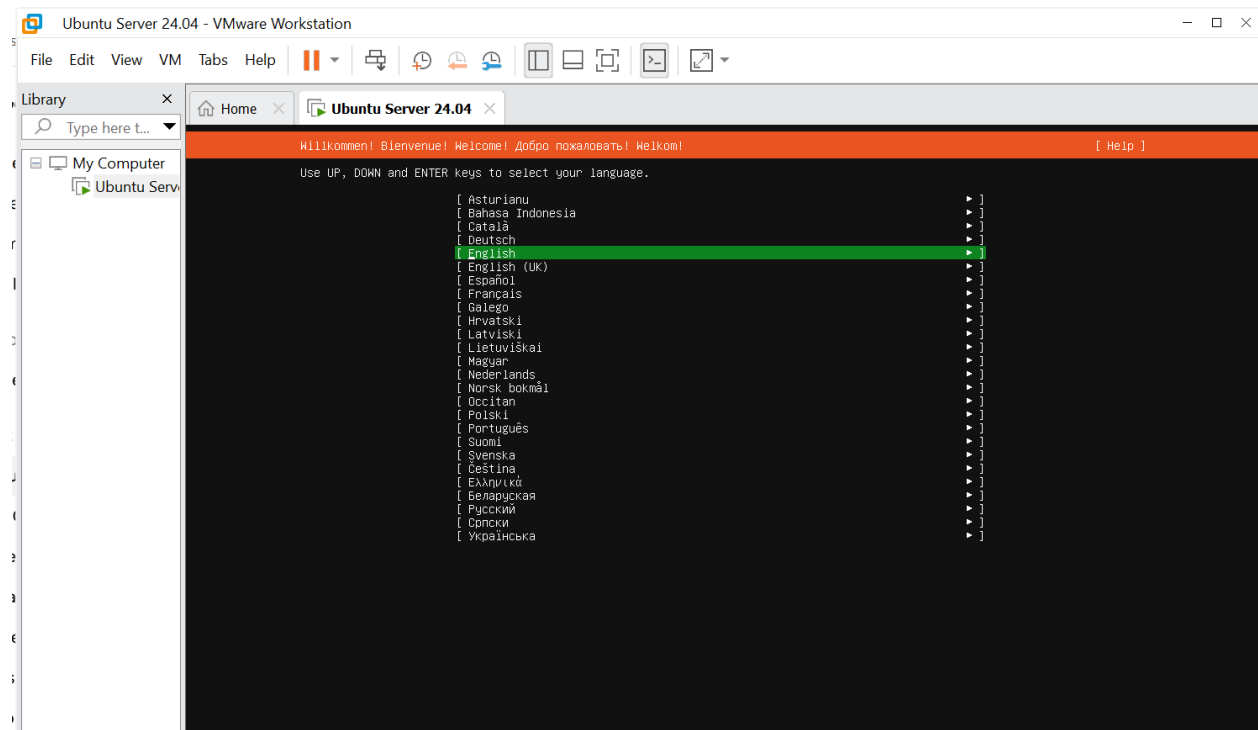
Figure 12: Final summary screen of the created Ubuntu server Virtual Machine hardware configuration.



Filename:ubuntu_installer_boot.png

Figure 13: Initial boot screen of Ubuntu Server installer inside VMware.

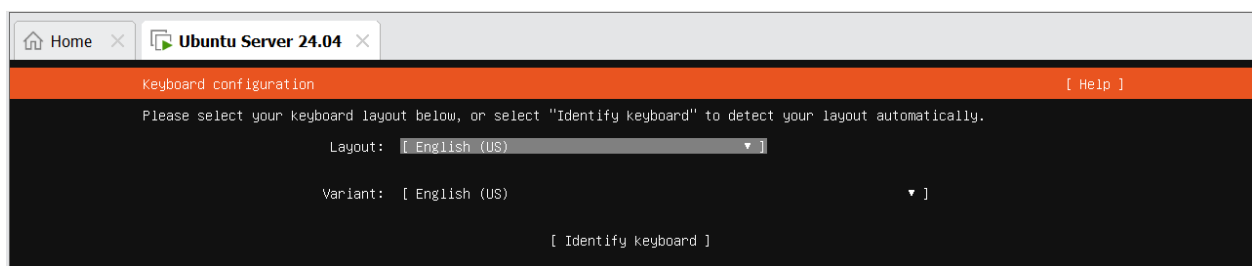
Task 5: Choose the Installation Language



Filename: ubuntu_Language.png

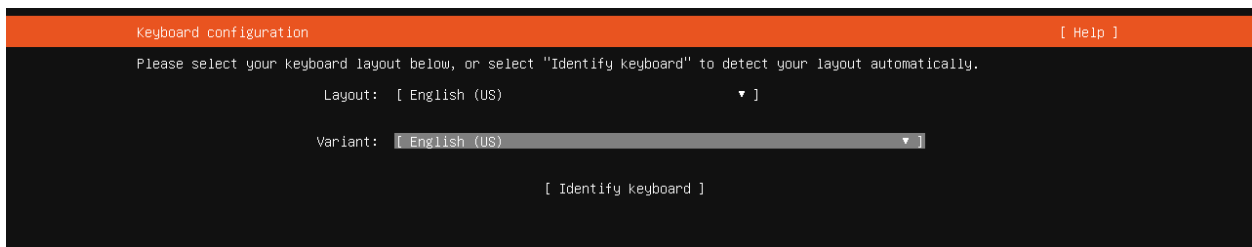
Figure 14: Selecting English as the preferred installation language.

Task 6: Configure the Keyboard Layout



Filename: ubuntu_keyboard_layout.png

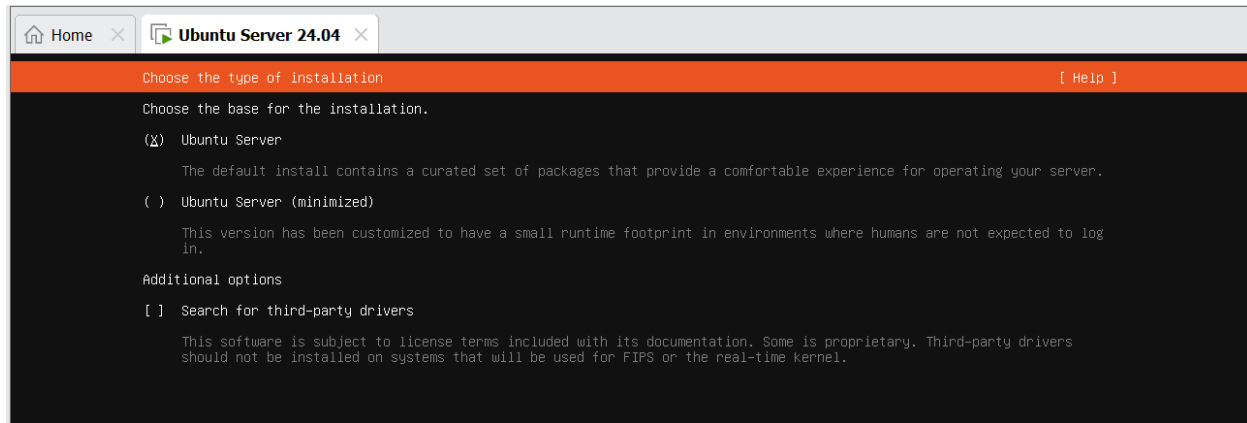
Figure 15 : Selecting the matching keyboard layout in the installer.



Filename: ubuntu_keyboard_variant.png

Figure 16 : Selecting the default keyboard variant.

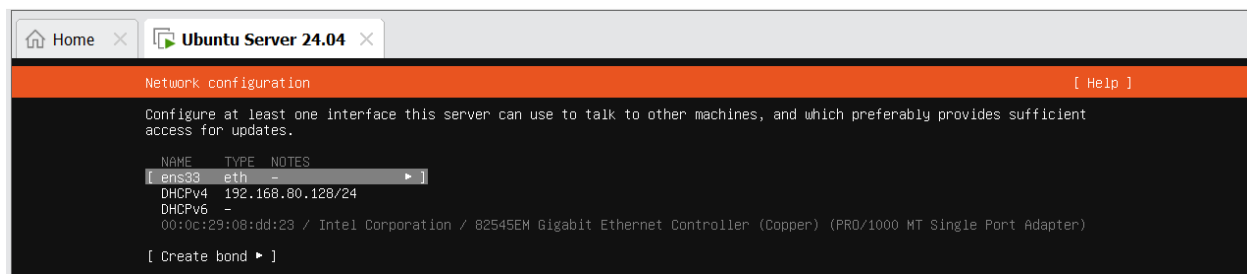
Task 7: Choose the Ubuntu Server Installation Type



Filename: ubuntu_installation _type.png

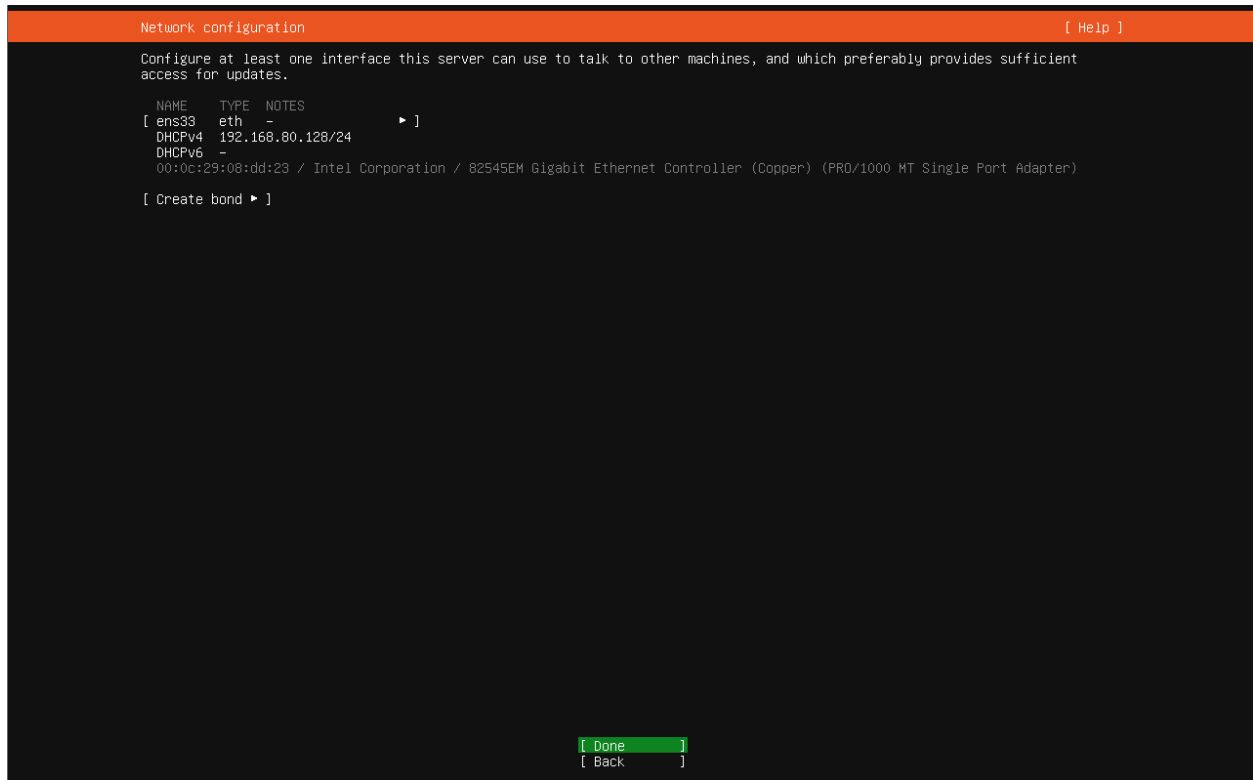
Figure 17: Selecting the base Ubuntu Server installation option.

Task 8: Configure the Network



Filename: ubuntu_network _interface.png

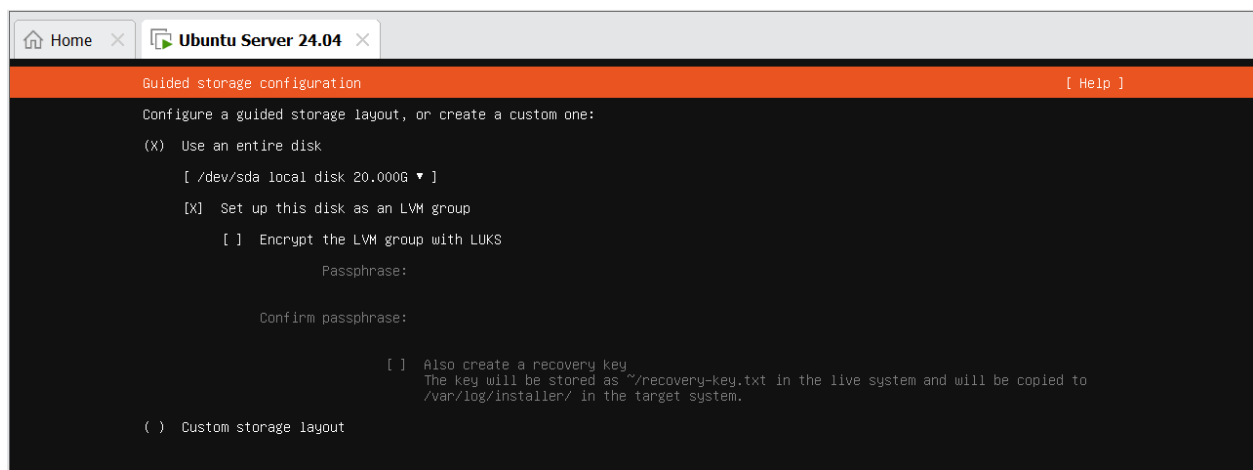
Figure 18 : Network interface detected automatically by the Ubuntu installer.



Filename: ubuntu_installer_network.png

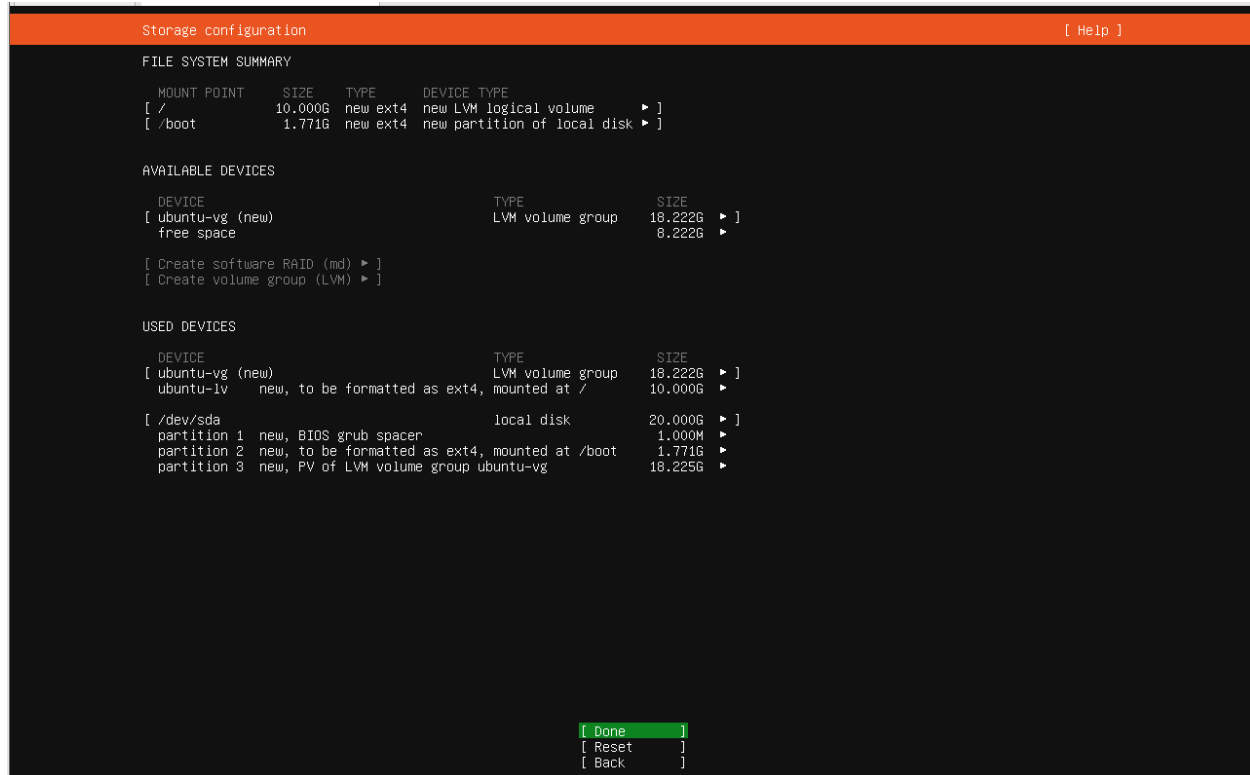
Figure 19: Verification of IPv4 address assigned to the server interface via DHCP.

Task 9: Configure Storage



Filename: ubuntu_guided_storage.png

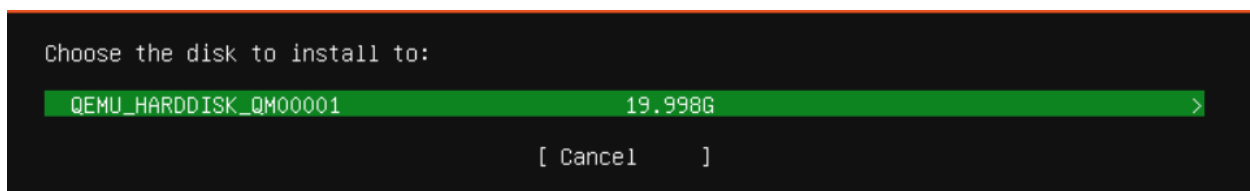
Figure 20: Selecting guided storage layout using the entire virtual disk.



Filename: ubuntu_storage_configuration.png

Figure 21: Reviewing the storage layout proposed by the installer.

Task 10: Select the Installation Disk



Filename: ubuntu_selected_disk.png

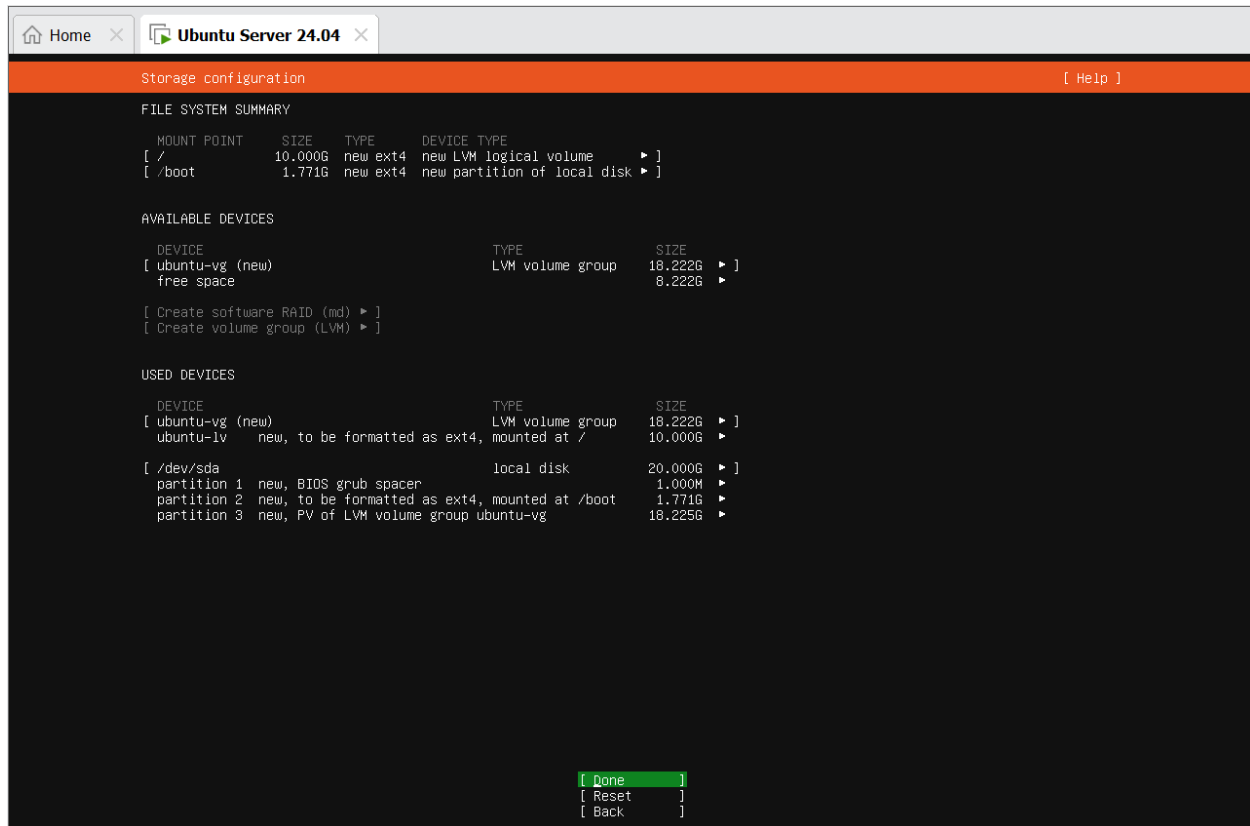
Figure 22: Target virtual disk selected for Ubuntu Server installation.



Filename: ubuntu_disk_details.png

Figure 23: Disk name and capacity details confirming the 20 GB VMware virtual disk.

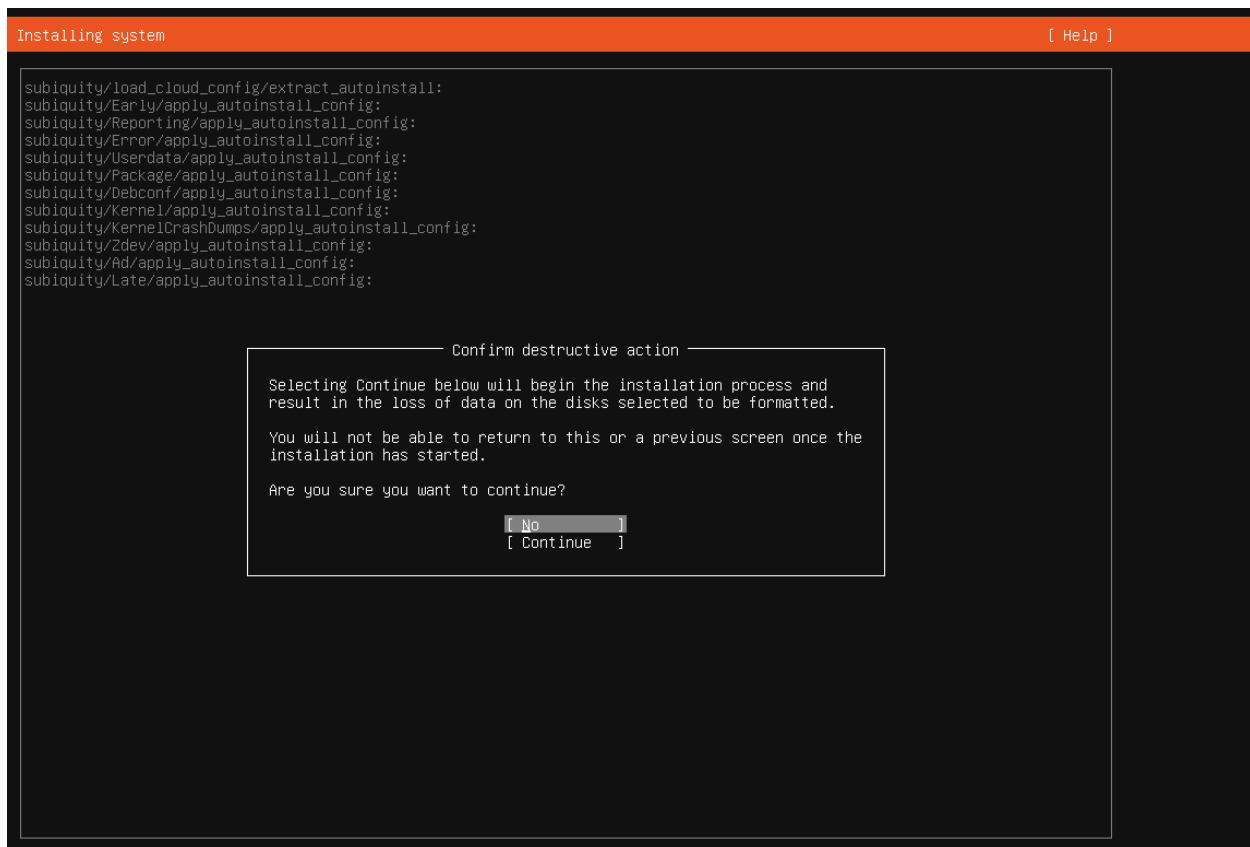
Task 11: Review the Partitions



Filename: ubuntu_partition_layout.png

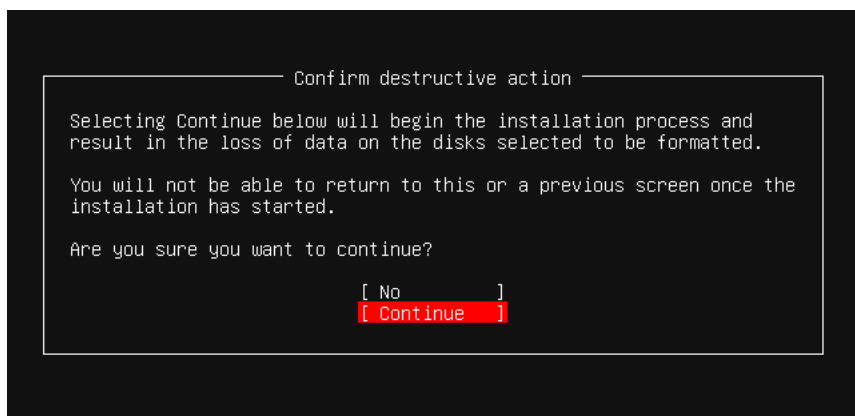
Figure 24 : Reviewing proposed system partition layout before writing changes.

Task 12: Confirm the Storage Changes



Filename: ubuntu_storage_warning.png

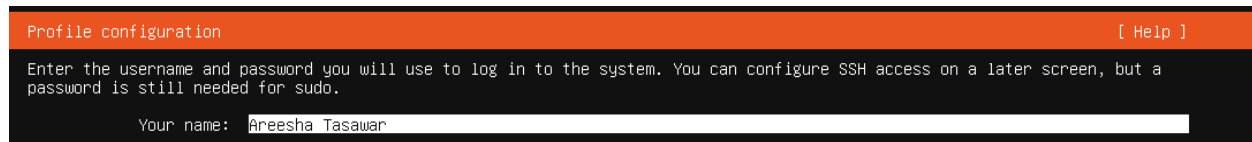
Figure 25: Destructive action prompt warning about disk formatting.



Filename: ubuntu_storage_confirmed.png

Figure 26 : Confirming and continuing with the partition changes.

Task 13: Configure the Ubuntu Profile



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Areesha Tasawar

Filename: ubuntu_actual_name.png

Figure 27: Profile setup screen showing full actual name entered.



Your servers name: ubuntu

The name it uses when it talks to other computers.

Filename: ubuntu_server_name.png

Figure 28: Server name configured as ubuntu.



Pick a username: areesha-tasawar

Filename: ubuntu_username.png

Figure 29 : Username field configured to match exact GitHub username.



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Areesha Tasawar

Your servers name: ubuntu

The name it uses when it talks to other computers.

Pick a username: areesha-tasawar

Choose a password: *****

Confirm your password: *****

Filename: ubuntu_profile_setup.png

Figure 30: Complete overview of filled profile configuration details.

Task 14: Complete the Software Installation

```
Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%2131287331' from host.

ubuntu login: areesha-tasawar
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 12 10:46:40 AM UTC 2026

System load:  0.02          Processes:            216
Usage of /:   50.5% of 9.75GB Users logged in:      0
Memory usage: 21%          IPv4 address for ens33: 192.168.80.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

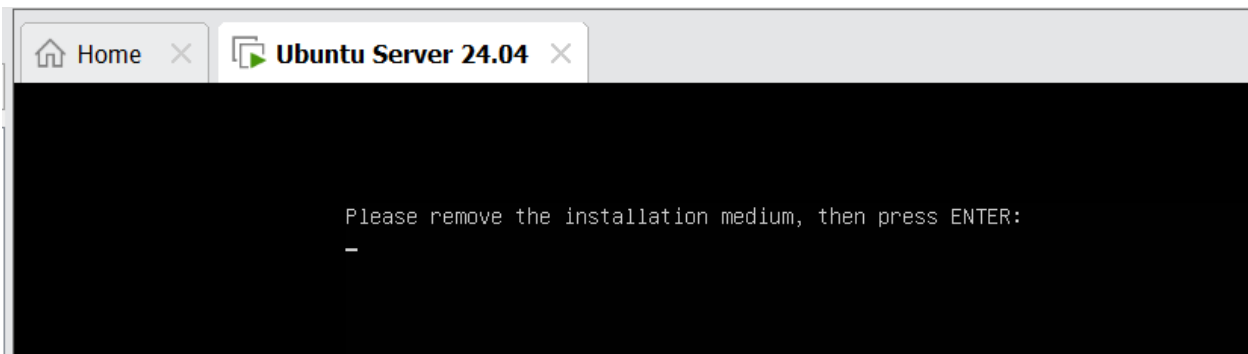
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Filename: ubuntu_installation_progress.png

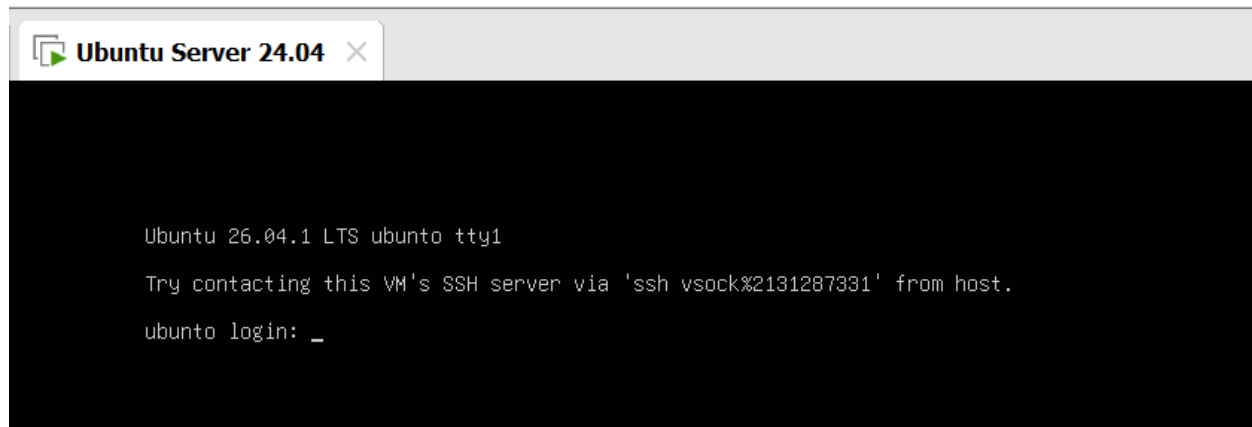
Figure 31: System installation and base package extraction in progress.

Task 15: Reboot and Verify the Ubuntu Identity



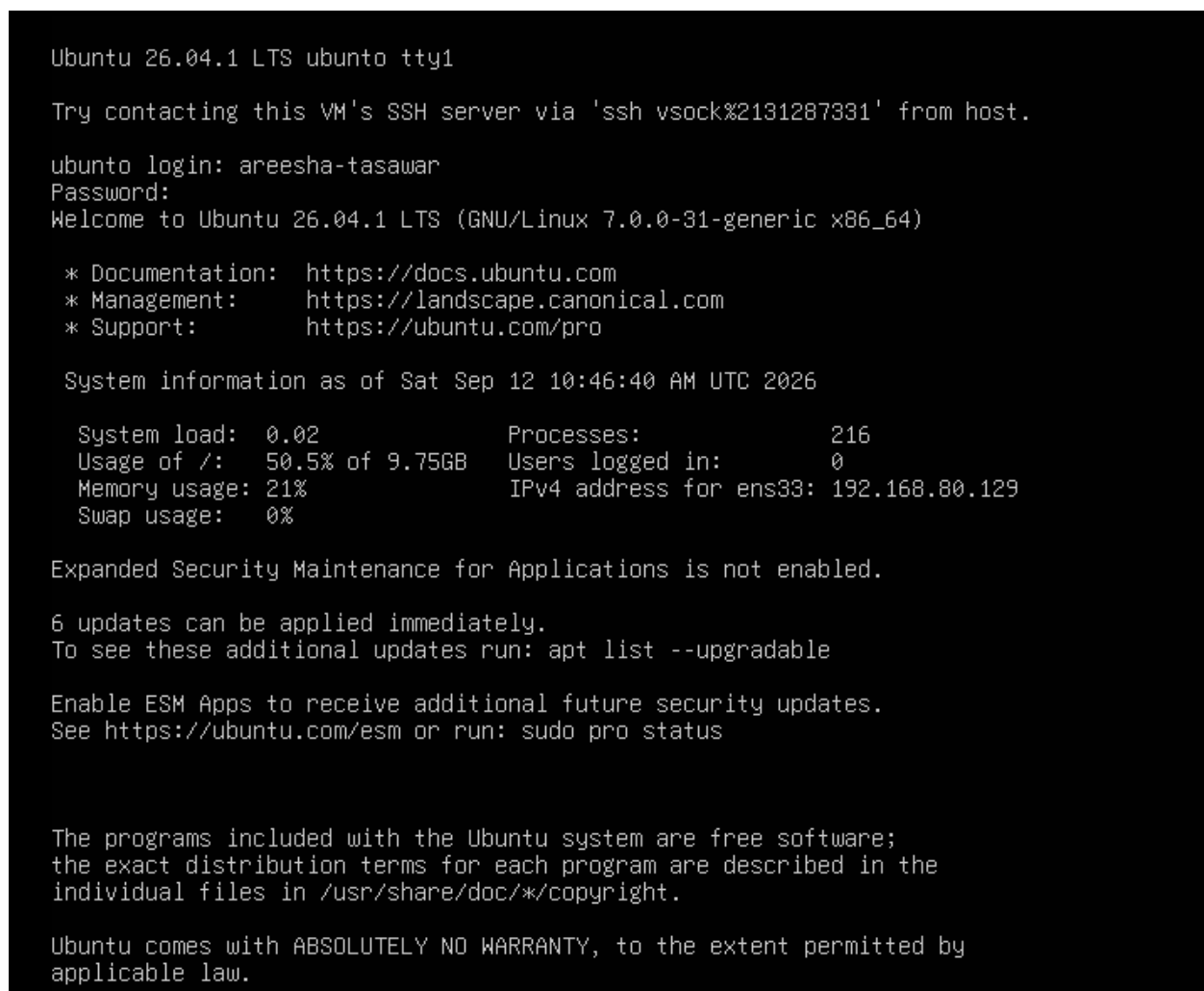
Filename: ubuntu_installation_complete.png

Figure 32: Installer screen reporting installation complete with Reboot Now option.



Filename: ubuntu_login_screen.png

Figure 33: Post-reboot TTY login prompt of Ubuntu Server.



Filename: ubuntu_terminal_login.png

Figure 34: Successful authentication into the Ubuntu Server console.

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

areesha-tasawar@ubuntu:~$ _
```

Filename: ubuntu_identity_verified.png

Figure 35: Terminal prompt verifying exact <github-username>@ubuntu:~\$ format.

Task 16: Find the Ubuntu Server IP Address

```
areesha-tasawar@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:08:dd:23 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2908dd23
    inet 192.168.80.129/24 metric 100 brd 192.168.80.255 scope global dynamic ens33
        valid_lft 959sec preferred_lft 959sec
    inet6 fe80::20c:29ff:fe08:dd23/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
areesha-tasawar@ubuntu:~$ _
```

Filename: ubuntu_ip_addr_command.png

Figure 36: Execution of ip addr command inside Ubuntu terminal.

```
    inet 192.168.80.129/24 metric 100 brd 192.168.80.255 scope global dynamic ens33
        valid_lft 959sec preferred_lft 959sec
    inet6 fe80::20c:29ff:fe08:dd23/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Filename: ubuntu_ip_address.png

Figure 37: Output highlighting active interface's IPv4 address.

Task 17: Connect to Ubuntu Server from Windows

```
C:\Users\User>ssh Areesha-Tasawar@192.168.80.129
The authenticity of host '192.168.80.129 (192.168.80.129)' can't be established.
ED25519 key fingerprint is SHA256:z1rLhJsFitmkvg20l50p6s/xP1ZbbsCGZbsajLySj54.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Filename: windows_ssh_command.png

Figure 38: Running ssh <username>@<ip> command in Windows Terminal.

```
THIS key IS NOT known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes|
```

Filename: windows_ssh_fingerprint.png

Figure 39: Accepting RSA key fingerprint prompt during initial SSH connection.

```
C:\Users\User>ssh areesha-tasawar@192.168.80.129  
areesha-tasawar@192.168.80.129's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Sat Sep 12 11:04:37 AM UTC 2026  
  
System load:  0.0                Processes:            221  
Usage of /:   50.8% of 9.75GB     Users logged in:     0  
Memory usage: 22%                IPv4 address for ens33: 192.168.80.129  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
6 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

Filename: windows_ssh_login.png

Figure 40: Successful remote terminal authentication into Ubuntu Server.

```
areesha-tasawar@ubuntu:~$ |
```

Filename: windows_ssh_identity.png

Figure 41: Remote SSH terminal displaying verified user identity prompt.